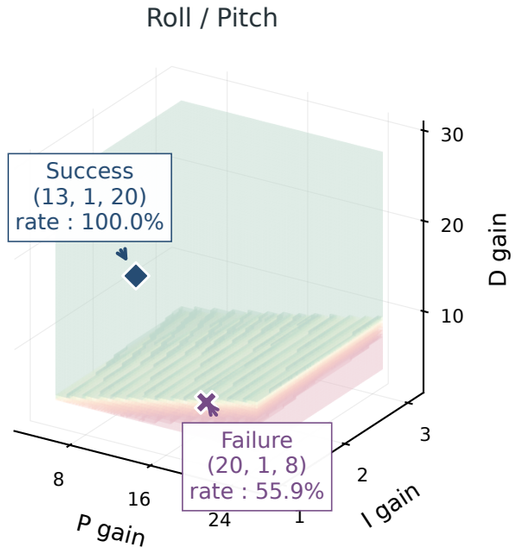
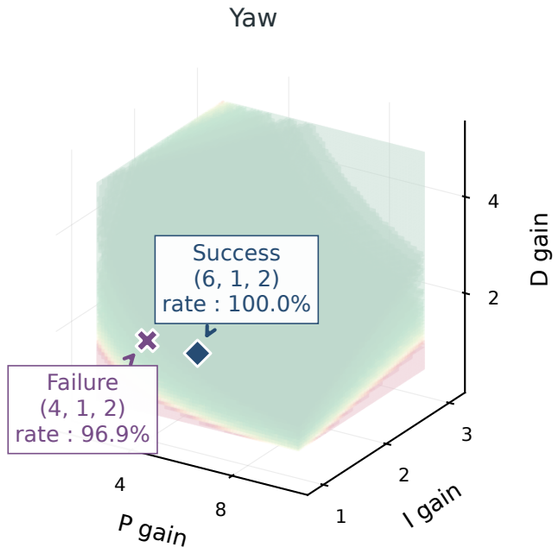
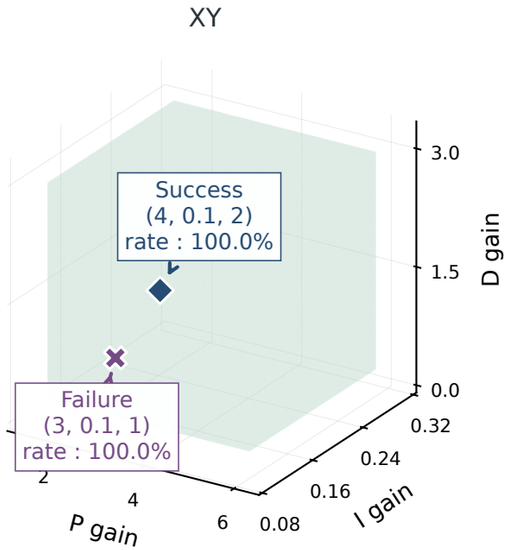
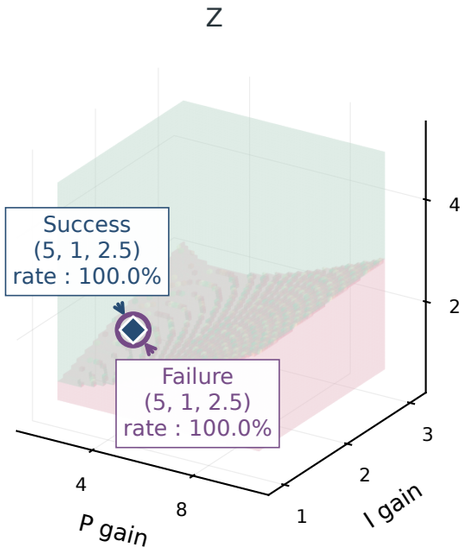
All PID gains: Success is green at 100% on every axis

Failure 1

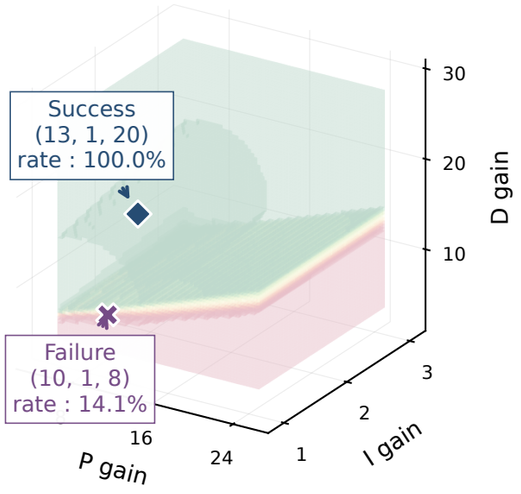


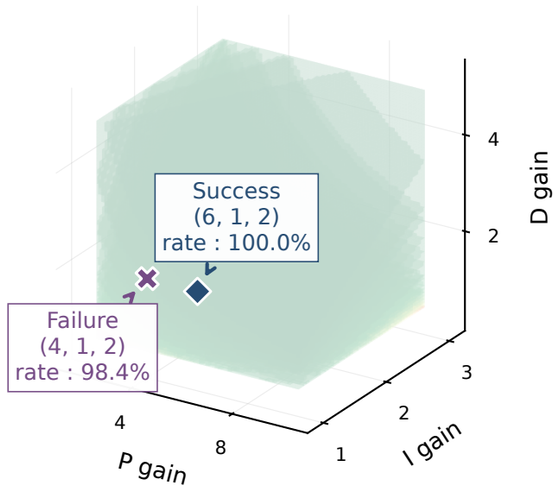


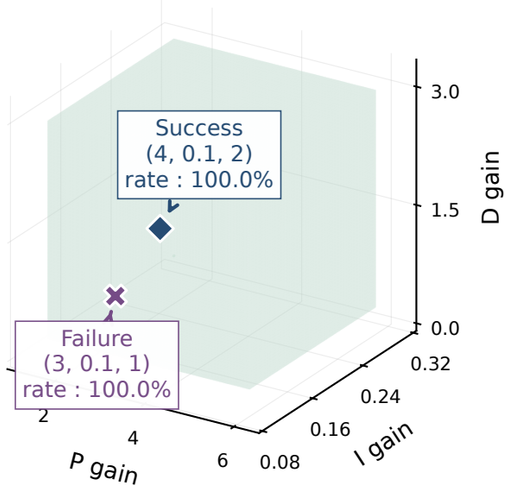


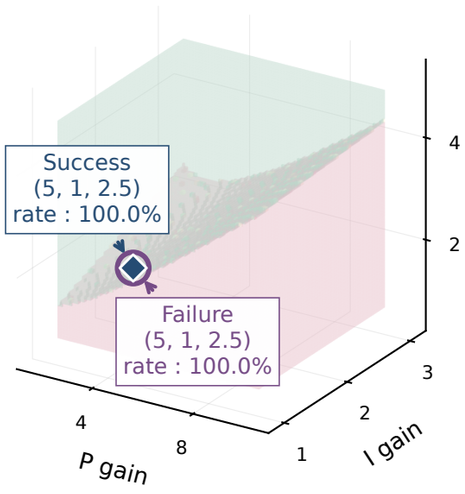


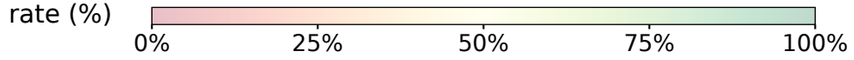
Failure 2











3D volumes are cut at each shared reference I to show both points. Purple marks identify the failure-flight gains; Z gains coincide.

Each case uses its own posterior. Other controller gains are fixed at the successful-flight values.